Musan (Hao) Wu

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Self-Introduction

My name is **Hao Wu**. I prefer to be called by my English name *Musan*, or just *Hao* (pronounced as "how"). I am currently a final-year Ph.D. student at the Institute of Software, University of Chinese Academy of Sciences (UCAS), where I am fortunate to be supervised by Prof. Naijun Zhan. I also completed my undergraduate studies in computer science at the same university.

Research Interests

My research interests primarily lie in two topics:

- **Formal Verification and Synthesis.** My goal is to advance extensible and scalable formal reasoning methodologies for verification and synthesis. I investigate a diverse range of systems exhibiting various computational behaviors, including discrete-time programs, continuous-time and hybrid dynamical systems, control systems, stochastic systems, and communication protocols.
- **Theoretical Computer Science.** I study foundational theoretical problems spanning logic, automata, algorithms, complexity, and computability, with the aim of deepening our understanding of these core concepts.

Education

- **Institute of Software, University of Chinese Academy of Sciences (UCAS)** — *Sep. 2020 - Present*
  Ph.D. in Computer Science at St. Key Lab. Comput. Sci. — Beijing, China
  - Advisor: Prof. Dr. Naijun Zhan
- **The Hong Kong University of Science and Technology (HKUST)** — *Mar. 2024 - May. 2024*
  PG Visiting Internship Students (Research) — Hong Kong, China
  - Advisor: Prof. Dr. Amir Goharshady
- **University of Chinese Academy of Sciences (UCAS)** — *Sep. 2016 - June. 2020*
  B.Sc. (with honor) in Computer Science. (Note:UCAS began to recruit undergraduate students in 2014) — Beijing, China
  - Advisor: Prof. Dr. Xiaoming Sun
  - GPA: 3.86/4. Distinguished B.Sc. Thesis Award.

Publications

Due to the doctoral graduation requirements in China, the author order for papers (primarily the *first* author) is determined *by contribution*, and we use this order as the default. Names marked with an asterisk (∗) denote *equal contribution*, and names marked with a dagger (†) indicate *alphabetical order*. PDF files can be found via my webpage.

1. **Hao Wu**, Qiuye Wang, Bai Xue, Naijun Zhan, Lihong Zhi, and Zhi-Hong Yang. 2025. **Synthesizing Invariants for Polynomial Programs by Semidefinite Programming**. ACM Trans. Program. Lang. Syst., Vol. 47(1): 1-35.
2. Xiong Xu, Jean-Pierre Talpin, Shuling Wang, **Hao Wu**, Bohua Zhan, Xinxin Liu, and Naijun Zhan. 2025. **HpC: A Calculus for Hybrid and Mobile Systems**. Proc. ACM Program. Lang., 9(OOPSLA1): 1158-1183.
3. Guanhua Lin, **Hao Wu**†, and Naijun Zhan. 2025. **Barrier Certificate Synthesis via Interpolation and Difference-of-Convex Programming**. In the Proc. of Colloquium on Design and Verification of Cyber-Physical Systems: From Theory to Applications, Festschrift of Prof. Martin Fränzle's 60th birthday.
4. **Hao Wu**, Yu-Fang Chen, Zhilin Wu, Bican Xia, and Naijun Zhan. 2024. **A decision procedure for string constraints with string/integer conversion and flat regular constraints**. Acta Informatica, 61(1):23–52.
5. **Hao Wu***, Jie Wang*, Bican Xia, Xiakun Li, Naijun Zhan, Ting Gan. 2024. **Nonlinear Craig Interpolant Generation Over Unbounded Domains by Separating Semialgebraic Sets**. In FM, 1: 92-110.
6. **Hao Wu**, Shenghua Feng, Ting Gan, Jie Wang, Bican Xia, and Naijun Zhan. 2024. **On Completeness of SDP-Based Barrier Certificate Synthesis over Unbounded Domains**. In FM, 2: 248-266.
7. Yihao Yin, **Hao Wu**, Shuling Wang, Xiong Xu, Fanjiang Xu, and Naijun Zhan. 2024. **The Design of Intelligent Temperature Control System of Smart House with MARS**. In SETTA . Springer, 217–235.
8. Shuling Wang, Bohua Zhan, Huanhuan Sheng, **Hao Wu**, Shicheng Yi, Lingtai Wang, Xiangyu Jin, Bai Xue, Jinghui Li, Shuang-Qing Xiang, Zhan Xiang, and Bifei Mao. 2022. **Survey on Requirements Classification and Formalization of Trustworthy Systems** (in Chinese). Journal of Software.
9. Xiaohui Bei, Xiaoming Sun, **Hao Wu**†, Jialin Zhang, Zhijie Zhang, and Wei Zi. 2020. **Cake Cutting on Graphs: A Discrete and Bounded Proportional Protocol**. In SODA, 2114–2123.

Academic Services

- **Teaching Assistant.**
  - 2023. Foundations of Theoretical Computer Science, B.Sc., UCAS
- **External Reviewer.**
  - Conferences: HSCC'25, FM'24, iFM'24, ISSAC'23, CDC'23, TACAS'23, ChinaSoft'23, ICTAC'22
  - Journals: TAC, Automatica, Journal of Symbolic Computation (JSC), TCAD.
- **Projects.**
  - Participant in National Key R&D Program of China under grant No. 2022YFA1005101
  - Participant in Natural Science Foundation of China under grant No. 62192732.

Additional Information

Languages: English (working language), Mandarin Chinese (native)